



CFD0013 · COMMUNICATION SKILLS 1

Title: Research Essay

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Academic Honesty Policy Statement

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The factors that impact Cyber Security to Cambodia's Banking sector

As the world becomes more digital, the race to protect financial institutions from cyber threats has never been more urgent. Every day, hackers find new ways to infiltrate systems, putting the security of banks and their customers at risk. Cybersecurity is no longer just a technical issue. It's a critical defense against potential financial disaster. In Cambodia, the banking sector faces unique challenges in staying secure. With more people using online banking and digital services, it's crucial that banks ensure their systems are protected against cyber-attacks. This requires the implementation of advanced technology and careful risk management. The factors that impact cybersecurity in Cambodia's banking sector include avoiding substantial financial losses, enhancing internet banking adoption under technology, implementing risk-based cybersecurity measures, and addressing the complexity of setup.

As Cambodia moves towards digitalization, all banks need to have a key role in protecting both personal and business financial assets and information. This shift to digital also makes the banking sector more exposed to cyberattacks, which can put sensitive data at risk and lead to serious consequences. Consequently, banks depend on customer trust to keep their money and information safe. To maintain this trust, experts stress the need for strong cybersecurity measures, including multi-factor authentication, regular software updates, encryption, and strict access controls. In this regard, Zokhir Rasulov, Chief Digital Officer at ABA Bank, points out the need for cybersecurity and data privacy measures that not only defend against attacks but also ensure smooth and convenient banking services. Additionally, all banks in Cambodia hold a lot of sensitive data, such as financial information, personal details, and transaction history, which is very appealing to cybercriminals. Kishore Kumar Muniswamy, Head of IT at Chief Commercial Bank Plc, notes that this data could be used for financial fraud, identity theft, and other harmful activities. Moreover, Niraj Gupta, Director of Macro Computing Solutions, also adds that cyberattacks can cause significant financial losses for banks through theft, fraud, or extortion. The costs of fixing breaches, legal fees, and regulatory fines can make the damage even worse. To address these risks effectively, Dinesh Elango, Associate Professor at the American University of Phnom Penh, advises additional measures such as encrypting data, conducting regular security audits, and educating employees and customers about cybersecurity knowledge. These steps are essential for preventing financial losses and protecting data privacy as Cambodia's banking sector continues to evolve. (Pazo, 2024)

The Cambodian banking sector has experienced significant growth in recent years, primarily driven by stable macroeconomic conditions and policies that encourage investment. One area that has particularly flourished is internet banking (IB), which gained substantial traction during the COVID-19 pandemic by offering customers the convenience of accessing financial services remotely. Given this backdrop, understanding the factors that influence mobile banking adoption has become crucial for optimizing its usage. This study aimed to explore consumer intentions to adopt IB by employing a questionnaire-based approach to collect and analyze quantitative data from 391 participants. The research applied the Technology Acceptance Model (TAM), integrating a subjective norm (SN) construct, to assess the behavior of Cambodian

consumers, utilizing PLS-SEM for data analysis. The findings underscored that trust and perceived usefulness are pivotal factors shaping attitudes toward IB and the intention to adopt it. Interestingly, perceived ease of use did not significantly influence adoption intentions. Moreover, the study emphasized the importance of socially influenced behaviors, or SN, in molding consumer attitudes, which subsequently impact IB adoption. Overall, the results offer both theoretical and empirical insights that contribute to a deeper understanding of the factors driving IB adoption in Cambodia. (Ly, 2022)

Establishing strong cybersecurity in Cambodia's banking sector is a complex challenge due to several factors. One major issue is the fast pace of digital changes in financial institutions, which often outstrips the speed at which necessary cybersecurity measures are developed. The mix of many outdated banking systems makes it even harder to create secure and connected networks. Additionally, there is a shortage of skilled cybersecurity professionals in Cambodia, limiting the expertise available to design and maintain these systems. Following international standards adds more complexity, as banks must navigate both local regulations and global cybersecurity requirements. Protecting banks from increasingly advanced cyber threats requires significant financial investment in the latest technology and staff training. Many banks might find it difficult to allocate enough resources, which can lead to vulnerabilities in their systems. The interconnected nature of banking networks means that a security breach in one bank could affect others, raising risks for everyone involved. In conclusion, outdated infrastructure, a lack of local expertise, and the growing sophistication of cyber threats make establishing effective cybersecurity in Cambodia's banking sector a challenging, long-term, and expensive task. This highlights the need for ongoing investment and cooperation between the public and private sectors to build strong systems capable of withstanding modern cyber threats. (Ness, 2023)

The cybersecurity measures in Cambodia, the NBC (National Bank Cambodia) Risk Management Guidelines cybersecurity have brought about a number of challenges, especially for small banks and financial institution. It can be difficult, costly, and overwhelming for small institutions to manage those guidelines, which call for intensive monitoring, visualizing, and compliance. The stringent regulations concerning IT security and access controls, although crucial, may also impede decision-making and make it more difficult for those establishment to adjust to market fluctuations and competition. In the current face-paced digital world, this might hinder their capacity to innovate and compete. Moreover, the demands on institutions with limited resources for highly skilled labor and cutting-edge technology can be problematic. This results in disparities in the level of cyber security protection that each organization receives, with smaller ones being particularly vulnerable. Furthermore, smaller institutions or organizations may find it difficult to finance the guidelines' substantial demands for trained personnel and technological investments. Due to their limited resources, smaller institutions are more vulnerable to cyber threats, which can result in disparities in the level of cyber security across the board. Since smaller financial institutions in Cambodia may find it more difficult to grow and remain stable because of these guidelines, even though their goals are to improve cybersecurity,

they may also widen the divide between better-funded, larger institutions and their less well-established counterparts. (National Bank of Cambodia, 2019)

In conclusion, as digital technology advances, safeguarding financial institutions against cyber threats has become increasingly critical. Hackers keep developing new methods to infiltrate systems, thereby endangering both banks and their customers. Cybersecurity has gone beyond its technical origins to become a vital shield against potential financial crises. The banking sector in Cambodia encounters unique challenges in maintaining security, particularly as online banking and digital services become more common. To protect against cyberattacks, banks must implement cutting-edge technology and manage risks with precision. The key factors influencing cybersecurity in Cambodia's banking sector include preventing substantial financial losses, promoting the adoption of internet banking, employing risk-based cybersecurity strategies, and addressing the complexities of system configurations.

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